

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. (CE) Examination****Apr-May 2018****CE245/CE201.02/CE201.01/CE201 Data Structures & Algorithms****Date: 30.04.2018, Monday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

Q - 1 Do as directed.

- (a) In a selection sort of n elements, how many times the swap function is called in the complete execution of the algorithm? **01**
- (a) 1 (b) $n - 1$
(c) $n \log n$ (d) n^2
- (b) Suppose we are sorting an array of eight integers using quicksort, and we have just finished the first partitioning with the array looking like this: 34 12 56 21 61 89 71 69 74. Which statement is correct? The pivot could be **01**
- (a) Either 61 or 89. (b) 61 but not 89.
(c) 89 but not 61. (d) Neither be 61 nor 89.
- (c) Which of the following is not a stable sorting algorithm in its typical implementation? **01**
- (a) Insertion Sort (b) Merge Sort (c) Quick Sort (d) Bubble Sort
- (d) For given array of float A [-100:52, 60:89] find the total number of elements. Assume that Base address is 5001. Find address of A [0, 71] element, if it is stored in row major order. **02**

- Q - 2 (a)** Find 32 element from below given data using linear search and binary search technique. Also note down total number of comparisons in both technique. **04**

5,8,11,15,25,28,29,32

- (b) Covert the following expression into its equivalent postfix expression using stack. **04**
- Show the contents of stack for each step.

$$a + b * (c^{(d-e)} * f) - g$$

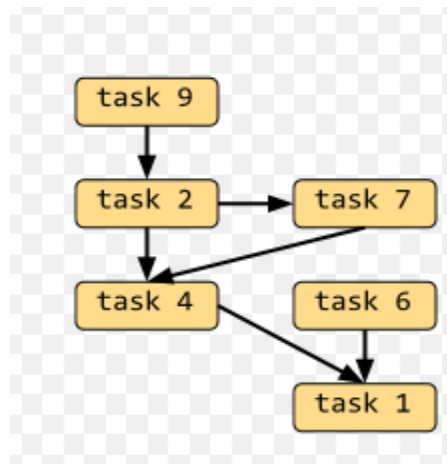
- (c) What are the advantages of circular queue over simple queue? Explain insertion operation in the Circular queue with all the conditions. **05**
- (d) For the given non empty doubly linked list, select correct steps for inserting element at front position. **02**

(L : Left most pointer, R : Right most pointer, new_node = pointer to new node)

- | | |
|---------------------------|---------------------------|
| (a) LPTR(new_node)= NULL; | (b) LPTR(new_node)= NULL; |
| RPTR(new_node)= L; | RPTR(new_node)= L; |
| LPTR(L) = new_node; | LPTR(R) = new_node; |
| L = new_node; | L = new_node; |
| (c) LPTR(new_node)= R; | (d) LPTR(new_node)= NULL; |
| RPTR(R)= new_node; | RPTR(new_node)= L; |
| RPTR(new_node) = NULL; | LPTR(L) = new_node; |
| R = new_node; | L = R |

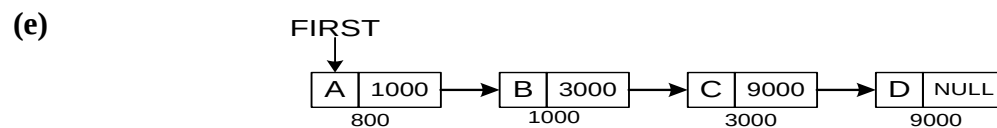
OR

- (d) Define topological ordering. Write all the possible Topological Ordering of the following graph. 02



Q:3 Answer following questions (Any Three) [15]

- (a) Show the sequence of integers (at the end of each pass) that represent the content of the array [25, 23, 15, 22, 6] after 4 passes of Insertion Sort.
- (b) A circular queue is given with the value of F=2, R=3 and size=3. Perform the following operations successively on the circular queue and show contents of Q, F and R after each operation. Insert (9), Insert (12), Delete, Delete, Delete, Insert (14), Insert (-4), Insert (1). (Initial value of queue is 10, 20)
- (c) Sort the following data using Radix Sort.
[COW, DOG, SEA, RUG, ROW, MOB, BOX, TAB, BAR]
- (d) Discuss tower of hanoi problem for N=3 discs with recursive tracing.



The following sequence of operations is performed on the above singly linked list. Show content of singly linked list after each operation.

Struct node *P;

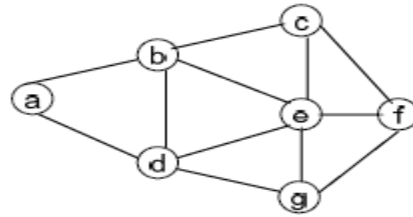
- (i) P = First → link; (ii) P → link → link → link =
 (iii) P = P → link → link → link; (iv) printf("%c", P → data);

SECTION – II

- Q - 4 (a)** Differentiate between array and linked list. 02
- (b) Which is the best data structure to check whether an arithmetic expression has balanced parenthesis or not? 01
- (c) Merge sort is not in-place Sorting Algorithm, Justify the statement. 02
- Q - 5 (a)** Evaluate the given expression with the help of stack. Show each step. 04
- 4 2 3 * + 5 8 / 2 * -**
- (b) The inorder and preorder traversal of a binary tree are **d b e a f c g** and **a b d e c f g**, respectively. Construct a tree and write down the postorder traversal of the binary tree. 03

- (c) Apply DFS on the following undirected graph (Starting Vertex is a)

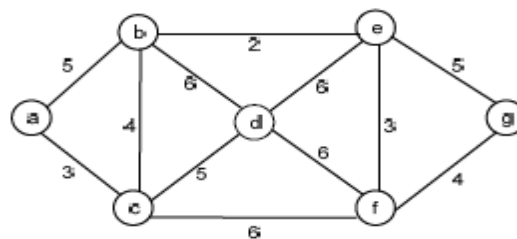
05



OR

- (c) Consider the following graph:

05



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm? Justify your answer.

- (A) (b, e) (e, f) (a, c) (b, c) (f, g) (c, d) (B) (b, e) (e, f) (a, c) (f, g) (b, c) (c, d)
 (C) (b, e) (a, c) (e, f) (b, c) (f, g) (c, d) (D) (b, e) (e, f) (b, c) (a, c) (f, g) (c, d)
- (d) The following sequence of operations is performed on a stack:
 PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20),
 POP. What is the sequence of values which are popped out?

03

Q - 6 Answer following questions (Any Three)

[15]

- (a) Explain Priority Queue. Also explain implementation methods for the Priority Queue.
- (b) A heap tree stores priorities (or priority-element) pairs at nodes. Demonstrate, step by step, the operation of Build-Min Heap on the array (Here only priorities of elements are given)
[103, 11, 101, 110, 111, 119, 19, 91]
- (c) Define graph. Explain different storage representation techniques of a graph.
- (d) Construct an AVL Tree from given inputs. Show each step with balance factor.
 Input Sequence : **21, 26, 30, 9, 4, 14, 28**
- (e) Arrange the following data in the array using hashing. Use linear probing to resolve collisions. Array size is 8 (index starts from 0)
 Data: **50, 47, 19, 21, 11, 45, 6, 25**
 Hash function **$H(x) = (x \bmod 5) + 3$**
